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Solid polymer/salt electrolytes based on linear poly((*N*-2-cyanoethyl)ethylenimine)

M. Erickson, R. Frech, D.T. Glatzhofer*

Department of Chemistry and Biochemistry, University of Oklahoma, 620 Parrington Oval, Norman, OK 73109, USA

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Abstract

There has been growing interest in the structure, properties, ionic association, and conductivity in poly(acrylonitrile) (PAN)/salt systems and its structural analogues. We have developed a simple procedure to cyanoethylate the nitrogen atoms ($>95\%$ by $^1\text{H-NMR}$) of linear poly(ethylenimine) (PEI) to form poly((*N*-2-cyanoethyl)ethylenimine) (PCEEI). IR studies indicate that, like PAN, a second nitrile stretching band develops for PCEEI on addition of LiTf, indicating coordination of nitrile groups to Li cations. However, spectroscopic data on the triflate ions indirectly suggest that, unlike PAN/LiTf systems, PCEEI systems contain significant proportions of “free” Li cations at high LiTf concentrations. Initial results indicate the RT ionic conductivity in the PCEEI systems is on the order of $10^{-8} \text{ S cm}^{-1}$ at a composition of N:Li = 6:1 and was found to be largely independent of LiTf concentration over a range of sample concentrations. The nature of the relative concentrations of ionic species of LiTf in PCEEI was determined with infrared spectroscopy. Results of room temperature and variable temperature complex impedance measurements are presented.

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Keywords: Poly(acrylonitrile); Poly(ethylenimine); Lithium triflate; Ionic association; Conductivity

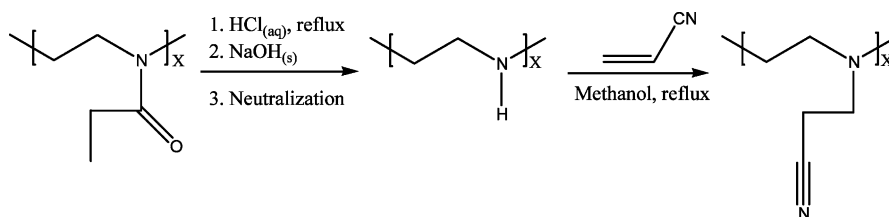
1. Introduction

As a result of their relatively high ionic conductivities and physical properties, polymer electrolytes have been studied for their application in solid thin-film batteries. Investigations of poly(ethylene oxide) (PEO), and poly(propylene oxide) (PPO), electrolyte systems have provided insight into the mechanism of ion transport and structure–property relationships [1–5]. Other polymeric electrolyte systems that have been developed in the pursuit of the same goals include poly(ethylenimine) (PEI), [6,7] and poly(acrylonitrile) (PAN) [8–13].

Linear poly(ethylenimine) (LPEI), is a structural analogue of PEO in which an NH replaces the oxygen atom in PEO, while PAN consists of a polyethylene backbone with a nitrile group on alternating carbons. Poly((*N*-2-cyanoethyl)ethylenimine) (PCEEI), is a previously unpublished hybrid of LPEI and PAN with a $-\text{CH}_2\text{CH}_2\text{CN}$ group attached to the nitrogen of the LPEI backbone (Scheme 1). Certain improvements upon the physical properties of PEI and PAN are evident. PAN has a higher glass transition temperature (80°C [13]) than PCEEI (-36°C [14]). PEI is highly crystalline until its melting point of $\sim 63^\circ\text{C}$ (onset temperature $\sim 52^\circ\text{C}$) [6], while PCEEI is not. In addition, preparation of PAN–electrolyte mixtures usually requires addition of plasticizer [8,9] or special hot pressing procedures [10–13] for salt (e.g. lithium triflate (LiTf)) dissolution, whereas PCEEI readily dissolves LiTf.

* Corresponding author. Tel.: +1-405-325-3834; fax: +1-405-325-6111.

E-mail address: dtglatzhofer@chemdept.chem.ou.edu (D.T. Glatzhofer).



Scheme 1. Synthesis of PCEEI from poly(2-ethyl-2-oxazoline).

The nature of the relative concentrations of ionic species of LiTf in PCEEI was determined with infrared spectroscopy. Results of room temperature and variable temperature complex impedance measurements are presented.

2. Experimental

2.1. Synthesis of poly((*N*-2-cyanoethyl)ethylenimine) (Scheme 1)

LPEI (3 g, 0.070 mol), synthesized as in reference [6], and methanol (35 ml) (Fisher Scientific) were added to a flask and allowed to dissolve with stirring. Upon dissolution, acrylonitrile (3.75 g, 0.071 mol, pure by NMR and IR) was added, the flask was fitted with a condenser and the reaction temperature was increased until solvent reflux. After 90 min from the onset of reflux, the reaction flask was removed from heat, and the solvent, as well as any excess acrylonitrile, was removed under reduced pressure leaving an amber-tinted, highly viscous liquid. PCEEI yield: 87%. Calculated ca. MW: $\sim 190\,000$ (assuming complete substitution and based on starting average MW for LPEI of 86 000 [6]). FT-IR (film cast from acetonitrile on KBr salt plate, cm^{-1}) 2245 ($\text{C}\equiv\text{N}$). ^1H NMR (CDCl_3 , ppm): 2.90–2.75 (q, $-\text{CH}_2\text{CN}$, br), 2.70–2.48 (s, $-\text{CH}_2\text{CH}_2\text{N}-$, br), 2.58–2.48 (t, $-\text{NCH}_2\text{CH}_2\text{CN}$, br). These are the actual multiplicities observed in the NMR spectrum obtained using a Varian 300 MHz NMR spectrometer. Selective irradiation, and consequential ^1H – ^1H decoupling, of each set of hydrogens on the side chain reveals that the protons on the carbon adjacent to the tertiary nitrogen are actually chemically nonequivalent. This is attributed to the chirality of the tertiary amine. The apparent quartet appears because each of these protons are split by each other and are also split by the two protons on the adjacent methylene group. The singlet resulting from the backbone protons has a shoulder that is indicative of protons associated with a small amount of unsubstituted repeat unit.

2.2. IR

PCEEI:LiTf samples were prepared by first dissolving PCEEI and LiTf (Aldrich) in anhydrous acetonitrile (Aldrich) in the desired N:Li (molar) ratios. Solutions of the PCEEI:LiTf samples were cast on KBr salt plates, and acetonitrile was evaporated over 48 h in a continuous purge dry box, leaving a thin polymer electrolyte film. Infrared spectra were taken at room temperature in an evacuated sample chamber using a Bruker IFS66V FT-IR spectrometer over a range of 4000–400 cm^{-1} at a resolution of 1 cm^{-1} . Curve fitting of the spectral data was done using commercially available Thermogalactic Grams AI 7.00 software. The spectra were curve-fit to a straight base line and one mixed Gaussian–Lorentzian product function for each band using a non-linear least squares method.

2.3. Complex impedance

Solutions of PCEEI and LiTf in acetonitrile were cast on 12.5 mm stainless steel electrodes, and acetonitrile was evaporated under vacuum over 96 h. Samples were compressed between a bottom electrode and a top electrode and placed in an air-tight cell. Sample temperatures were controlled using a circulating bath. Complex impedance measurements were made over the range of 5 Hz to 10 MHz using an HP 4192A LF impedance analyzer. Impedance plots were fitted using LEVM version 7.1.

3. Results and discussion

3.1. Infrared spectroscopy

The $\nu_s(\text{SO}_3)$ and $\delta_s(\text{CF}_3)$ vibrational modes of the triflate anion, CF_3SO_3^- , allow for the determination of the distribution of triflate species (CF_3SO_3^- , LiCF_3SO_3 , $\text{Li}_2\text{CF}_3\text{SO}_3^+$, etc.) in solution [15]. As seen in PEO, each triflate species has a characteristic $\nu_s(\text{SO}_3)$ and $\delta_s(\text{CF}_3)$ vibrational mode, with more highly associated triflate species shifted to higher frequencies [15]. Examination of these vibrational modes allows for estimation of the relative concentrations of species present as a function of salt concentration. Fig. 1 shows spectral bands for the

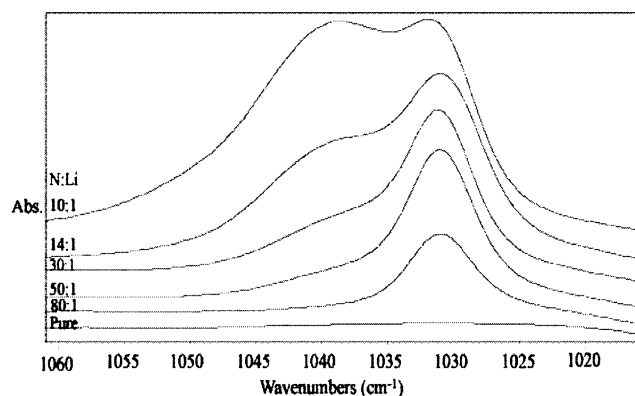


Fig. 1. IR spectra (cm^{-1}) of PCEEI:LiTf at 10:1, 14:1, 30:1, 50:1 and 80:1 compositions and pure PCEEI in the $\nu_s(\text{SO}_3)$ region.

$\nu_s(\text{SO}_3)$ mode of PCEEI:LiTf samples at nitrogen:lithium ratios of 10–80:1 with pure PCEEI as a reference. The spectral band at 1031 cm^{-1} for the 80:1 sample is attributable to non-coordinated, “free” triflate ion [15], while, at higher salt concentration, a second band at 1039 cm^{-1} increases in relative intensity and is attributed to contact ion-pair formation [15]. As can be seen, this band becomes approximately equal in intensity to the “free” ion band.

These results were quantified by determining the integrated band intensity of each species present by curve-fitting (Table 1). In contrast to PAN:LiTf electrolytes, where “free” triflate or lithium triflate contact ion-pairs are not observed at concentrations from N:Li = 10 to 60:1 [11], “free” anion in PCEEI:LiTf electrolytes is the dominant species at low salt concentrations, 30–80:1, while at a 10:1 salt concentration, the dominant species present shifts to contact ion-pair. The band that develops at 1053 cm^{-1} is attributed to the triple ion species [15]. Lithium triflate species in PCEEI appear to be more similar in the nature and degree of ion association to lithium triflate species in PEI, where the dominant species present, as determined by curve fitting, was also contact ion-pair at a 10:1 salt concentration [6]. As in PCEEI, lithium triflate aggregation increases with concentration in the PEI:LiTf electrolytes, but at a 5:1 concentration, the $(\text{Li}_2\text{Tf})^+$ species dominates in the absence of “free” triflate [6]. Examination of the $\delta_s(\text{CF}_3)$ vibrational mode in PCEEI:LiTf samples reveals com-

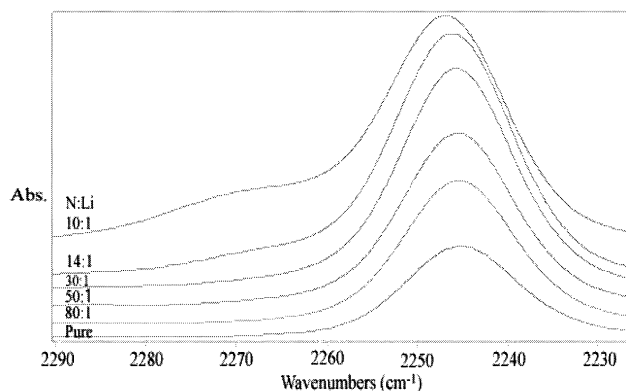


Fig. 2. IR spectra (cm^{-1}) of PCEEI:LiTf at 10:1, 14:1, 30:1, 50:1 and 80:1 compositions and pure PCEEI in the $\nu_s(\text{CN})$ region.

parable speciation behavior but is complicated due to a polymer band in this region.

It has been demonstrated by several groups that a high frequency shoulder on the $\nu_s(\text{C}\equiv\text{N})$ band is present in PAN:LiTf samples at high LiTf concentrations and has been attributed to nitrile complexation with Li^+ [8,9,12]. At sufficiently large salt concentration, this shoulder has been reported to be higher in relative intensity than the free nitrile band at 2244 cm^{-1} [12]. At salt concentrations of 10:1 and higher, a higher frequency shoulder on the PCEEI $\nu_s(\text{C}\equiv\text{N})$ band develops and increases in relative intensity with increasing salt coordination (Fig. 2). This implies that, at sufficient salt concentration, lithium complexation to the backbone nitrogen is not the sole source of polymer–lithium interaction.

Two potential Lewis base sites exist in a repeat unit of PCEEI: the nitrogen of the tertiary amine and the cyano nitrogen. Qualitative determination of the extent of lithium coordination to either site requires a comparison of vibrational modes primarily involving the tertiary amine nitrogen and modes primarily involving the cyano nitrogen. Data collected at present has been inconclusive and other studies, such as using Raman spectroscopy to examine the $\nu_s(\text{C}-\text{N})$ vibrational mode, are necessary.

3.2. Ionic conductivity

Room temperature impedance measurements on PCEEI:LiTf samples of 6, 20, 40 and 80:1 salt concen-

Table 1

Curve-fitting results showing the percentage of integrated band intensity (in parenthesis) for PCEEI:LiTf in the $\nu_s(\text{SO}_3)$ spectral region

Composition, N:Li	Band center frequency (cm^{-1}) (intensity)		
80:1			1031 (99%)
50:1		1039 (23%)	1031 (77%)
30:1	1053 (8%)	1039 (34%)	1031 (65%)
14:1	1052 (8%)	1039 (50%)	1031 (50%)
10:1	1053 (2%)	1039 (81%)	1031 (16%)

Table 2

Room temperature conductivity values for 6:1, 20:1, 40:1 and 80:1 compositions

Composition, N:Li	σ (S cm ⁻¹)
6:1	4.25×10^{-8}
20:1	4.45×10^{-8}
40:1	7.45×10^{-8}
80:1	9.15×10^{-8}

trations were performed, yielding conductivities on the order of 10^{-8} S cm⁻¹ (Table 2). An appreciable change in the conductivity of PCEEI:LiTf had been expected because the nature of the ionic species changes from predominantly “free” ion to predominantly contact ion-pair over a significant composition range. The contrary results may indicate morphological changes that offset conductivity changes due to speciation. Results of impedance measurements in the temperature range of 283–333 K for a 10:1 sample indicate a three order of magnitude increase of conductivity increasing with temperature 1.45×10^{-5} S cm⁻¹ at 60 °C. An Arrhenius plot of this data (Fig. 3) gives a calculated apparent activation energy value of 100 kJ mol⁻¹. Forsyth et al. reported activation energies of 55–140 kJ mol⁻¹ determined in the temperature range of 313–383 K in samples of PAN:LiTf at high salt concentrations (33–67% salt by weight) [13]. The value of 100 kJ mol⁻¹ found in PCEEI:LiTf is in the middle of the range reported by Forsyth et al. and is significantly higher than that in PEO:LiTf [16].

The behavior of PCEEI:LiTf stands in contrast to PAN:LiTf systems which, at 65 °C, show a dramatic

increase in conductivity from 10^{-9} to 10^{-6} S cm⁻¹ with increasing lithium triflate concentration from 50 to 75 wt.% [10]. However, at 65 °C, PAN:LiTf systems are below the Tg of the host polymer and at 60 °C, PCEEI:LiTf is well above Tg for the host. More extensive studies of the speciation and conductivity behavior of PCEEI:LiTf as a function of both temperature and LiTf concentration are ongoing to separate speciation from morphological effects and to understand its behavior relative to PAN:LiTf systems.

4. Conclusions

PCEEI is a new polymer synthesized by a simple Michael reaction of PEI with acrylonitrile. The reaction makes use of the functionalizable secondary amine in the backbone of PEI and represents one of the many potential modifications of PEI. While in PAN:LiTf samples no “free” ion exists from low to high salt concentration, “free” ion is observed in PCEEI:LiTf from 10 to 80:1 sample concentrations. Lithium coordination to the cyano functionality in PCEEI, as seen in PAN, is noted at 10 and 14:1 sample concentrations. Room temperature conductivity in the composition range of 6–80:1 is relatively concentration independent and is unusual considering the shift in LiTf speciation from essentially all “free” ion at a composition of 80:1 to predominantly contact ion-pair at 6:1. To achieve a better understanding of the mechanism for conduction in PCEEI:LiTf samples, it will be necessary to extend the comparative conductivity/spectroscopic studies to a wider polymer–salt composition range and over a larger temperature range. This is the goal of further studies of this unusual polymer electrolyte planned by the authors.

References

- [1] C. Berthier, W. Gorecki, M. Minier, M.B. Armand, J.M. Chabagno, P. Rigaud, *Solid State Ionics* 11 (1983) 91.
- [2] B.L. Papke, M.A. Ratner, D.F. Shriver, *J. Electrochem. Soc.* 129 (1982) 1434.
- [3] G. Peterson, P. Jacobsson, L.M. Torell, *Electrochim. Acta* 37 (1992) 1495.
- [4] M.E. Ries, M.G. Bereton, J.M. Cruickshank, P.G. Klein, I.M. Ward, *Macromolecules* 28 (1995) 3282.
- [5] R. Frech, S. Chintapalli, P.G. Bruce, C.A. Vincent, *Chem. Commun.* (1997) 157.
- [6] S. York, R. Frech, A. Snow, D. Glatzhofer, *Electrochim. Acta* 46 (2001) 1533.
- [7] R. Tanaka, T. Fujita, N. Nishibayashi, S. Saito, *Solid State Ionics* 60 (1993) 119.
- [8] S.R. Starkey, R. Frech, *Electrochim. Acta* 42 (1996) 471.
- [9] Z. Wang, B. Huang, H. Huang, L. Chen, R. Wue, F. Wang, *Electrochim. Acta* 41 (1996) 1443.

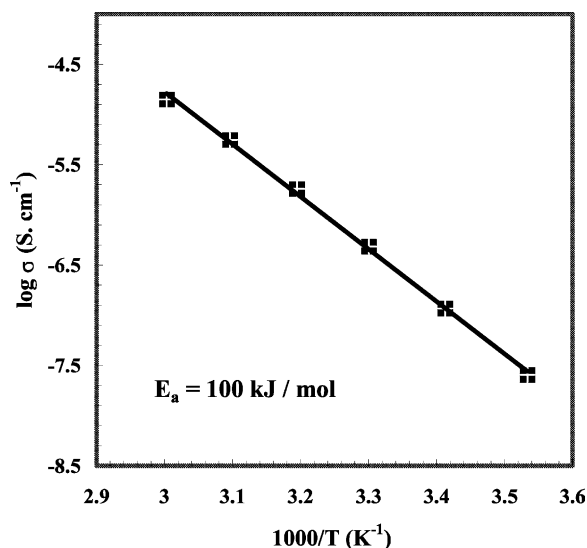


Fig. 3. Arrhenius plot of the $\log_{10}$ of conductivities vs. $1000/T$ for the 10:1 PCEEI:LiTf sample.

- [10] M. Forsyth, J. Sun, D.R. MacFarlane, A.J. Hill, *J. Polym. Sci.: Part B: Polym. Phys.* 38 (2000) 341.
- [11] A. Ferry, L. Edman, M. Forsyth, D.R. MacFarlane, J. Sun, *J. Appl. Phys.* 86 (1999) 2346.
- [12] A. Ferry, L. Edman, M. Forsyth, D.R. MacFarlane, J. Sun, *Electrochim. Acta* 45 (2000) 1237.
- [13] M. Forsyth, J. Sun, D.R. MacFarlane, *Solid State Ionics* 112 (1998) 161.
- [14] M. Erickson, R. Frech, D.T. Glatzhofer, unpublished results.
- [15] R. Frech, W. Huang, M.A.K.L. Dissanayake, *Mater. Res. Symp. Proc.* 369 (1995) 523.
- [16] J.R.M. Giles, *Solid State Ionics* 24 (1987) 155.